Prof. Dr. Luis Filipe Alves Pereira
Curso: Aprendizagem de Máquina
2026

UFAPE

eXplainable ARTIFICIAL INTELLIGENCE

Machine Learning Interpretability
Termo de pesquisa

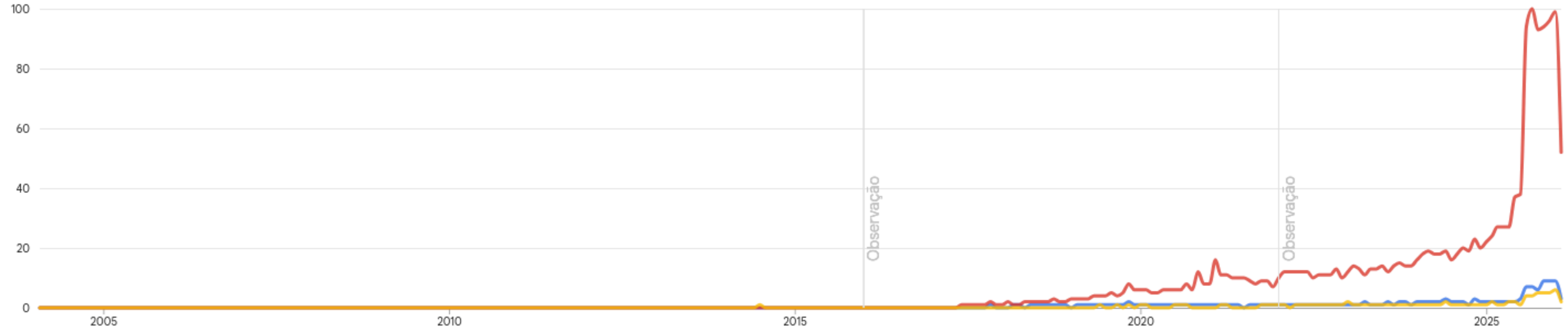
explainable AI
Termo de pesquisa

Machine Learning Explainability
Termo de pesquisa

Global | 1 de fev. de 2004 - 14 de fev. de 2026 | Pesquisa Google na Web

Interesse ao longo do tempo

Global · 1 de fev. de 2004 - 14 de fev. de 2026



U.S. FOOD & DRUG ADMINISTRATION | Health Canada Santé Canada | Medicines & Healthcare products Regulatory Agency

Transparency for Machine Learning-Enabled Medical Devices: Guiding Principles

June 2024

In 2021, the U.S. Food and Drug Administration (FDA), Health Canada and the United Kingdom's Medicines and Healthcare products Regulatory Agency (MHRA) jointly identified 10 guiding principles for good machine learning practice (GMLP). GMLP supports the development of safe, effective and high-quality artificial intelligence/machine learning technologies that can learn from real-world use and, in some cases, improve device performance.

Learn more: Good machine learning practice for medical device development: Guiding principles

The FDA, Health Canada and MHRA have further identified guiding principles for transparency for machine learning-enabled medical devices (MLMDs). These principles build upon the GMLP principles, especially:

- principle 7: Focus is placed on the performance of the human-AI team.
- principle 9: Users are provided clear, essential information.

While the guiding principles presented here promote transparency for MLMDs, transparency is a good practice to consider for all medical devices.

In this document, "transparency" describes the degree to which appropriate information about a MLMD (including its intended use, development, performance and, when available, logic) is clearly communicated to relevant audiences. "Logic" refers to information about how an output or result was reached or the basis for a decision or action. The degree to which this logic can be explained in a way that a person can understand is known as "explainability". Logic and explainability are aspects of transparency.

It's also good practice to explain how the device fits in the health care workflow. This would includ
description of the intended inputs and outputs. It would also describe how the output is intended to imp
health care decisions or actions, or the judgment of a health care professional. Details about dev
performance, benefits and risks can help one decide how best to use the device. Information about
management activities across its lifecycle, such as bias management strategies, can also be helpful.

It's valuable to provide the basis of a device output or other information that explains how the MLMD reach

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its output (the "logic" of the model), when this information is available and easily understood. Access to t
information helps one critically assess the device and its output when patient care decisions are being ma
Another good practice is to provide information on product development and risk management activit
across the total product lifecycle. Relevant information:

- explains the underlying technology and machine learning approaches.
- characterizes the training and testing data.
- provides summaries of clinical studies.
- provides ongoing updates on:
  - model and dataset characteristics.
  - performance monitoring.
  - detection and investigation of issues and risks across the total product lifecycle.

https://www.fda.gov/media/179269/download?attachment

BRIEFING

EU Legislation in Progress

European Parliament

Artificial intelligence act

OVERVIEW

European Union lawmakers signed the artificial intelligence (AI) act in June 2024. The AI act, the first binding worldwide horizontal regulation on AI, sets a common framework for the use and supply of AI systems in the EU.

The new act offers a classification for AI systems with different requirements and obligations tailored to a 'risk-based approach'. Some AI systems presenting 'unacceptable' risks are prohibited. A wide range of 'high-risk' AI systems that can have a detrimental impact on people's health, safety or on their fundamental rights are authorised, but subject to a set of requirements and obligations to gain access to the EU market. AI systems posing limited risks because of their lack of transparency will be subject to information and transparency requirements, while AI systems presenting only minimal risk for people will not be subject to further obligations.

The regulation also lays down specific rules for general purpose AI (GPAI) models and lays down more stringent requirements for GPAI models with 'high-impact capabilities' that could pose a systemic risk and have a significant impact on the internal market.

The AI Act was published in the EU's Official Journal on 12 July 2024. It entered into force in August 2024.

EU action,[3] as well as two legislative resolutions asking the Commission to establish a legal framework of ethical principles for the development, deployment and use of AI, robotics and related technologies in the Union and harmonising the legal framework for civil liability claims and imposition of a regime of strict liability on operators of high-risk AI systems.

In the past, the Council has repeatedly called for the adoption of common AI rules, including in 2017 and 2019. In 2020, the Council called upon the Commission to put forward concrete proposals that take existing legislation into account and follow a risk-based, proportionate and, if necessary, regulatory approach.

The Commission launched a broad public consultation in 2020 and published an Impact Assessment of the regulation on artificial intelligence, a supporting study and a draft proposal, which received feedback from stakeholders.[4] In its impact assessment, the Commission identified several problems raised by the development and use of AI systems, due to their **specific characteristics**, namely: (i) opacity (limited ability of the human mind to understand how certain AI systems operate), (ii) complexity, (iii) continuous adaptation and unpredictability, (iv) autonomous behaviour, and (v) functional dependence on data and on the quality of data.

An increasing number of countries worldwide are designing and implementing AI governance legislation and policies. While the United States (US) had initially taken a lenient approach towards AI, calls for regulation have recently been mounting. The White House has released the Blueprint for an AI Bill of Rights, a set of guidelines to protect the rights of the American public in the age of AI and President Joe Biden signed an executive order on AI in 2023. Chinese government agencies approved some guidelines on generative AI, while the UK has announced a pro-innovation approach to AI regulation, which largely regulates AI via existing laws. At international level, the Organisation for Economic Co-operation and Development (OECD) adopted some non-binding Principles on AI, in 2019, UNESCO embraced a set of Recommendations on the Ethics of AI in 2021, the G7 agreed some International Guiding Principles on Artificial Intelligence in 2023 and the Council of Europe adopted an international convention on AI in May 2024. Furthermore, in the context of the newly established EU-US tech partnership (the Trade and Technology Council), the EU and the US are seeking to develop a mutual understanding on the principles underpinning trustworthy and responsible AI.

PROJETO DE LEI Nº , DE 2023

Dispõe sobre o uso da Inteligência Artificial.

O CONGRESSO NACIONAL decreta:

CAPÍTULO I
DISPOSIÇÕES PRELIMINARES

Art. 1º Esta Lei estabelece normas gerais de caráter nacional para o desenvolvimento, implementação e uso responsável de sistemas de inteligência artificial (IA) no Brasil, com o objetivo de proteger os direitos fundamentais e garantir a implementação de sistemas seguros e confiáveis, em benefício da pessoa humana, do regime democrático e do desenvolvimento científico e tecnológico.

Art. 2º O desenvolvimento, a implementação e o uso de sistemas de inteligência artificial no Brasil têm como fundamentos:

I – a centralidade da pessoa humana;

II – o respeito aos direitos humanos e aos valores democráticos;

III – o livre desenvolvimento da personalidade;

IV – a proteção ao meio ambiente e o desenvolvimento sustentável;

V – a igualdade, a não discriminação, a pluralidade e o respeito aos direitos trabalhistas;

VI – o desenvolvimento tecnológico e a inovação;

VII – a livre iniciativa, a livre concorrência e a defesa do consumidor;

CAPÍTULO II
DOS DIREITOS

Seção I
Disposições Gerais

Art. 5º Pessoas afetadas por sistemas de inteligência artificial têm os seguintes direitos, a serem exercidos na forma e nas condições descritas neste Capítulo:

I – direito à informação prévia quanto às suas interações com sistemas de inteligência artificial;

II – direito à explicação sobre a decisão, recomendação ou previsão tomada por sistemas de inteligência artificial;

III – direito de contestar decisões ou previsões de sistemas de inteligência artificial que produzam efeitos jurídicos ou que impactem de maneira significativa os interesses do afetado;

IV – direito à determinação e à participação humana em decisões de sistemas de inteligência artificial, levando-se em conta o contexto e o estado da arte do desenvolvimento tecnológico;

V – direito à não-discriminação e à correção de vieses discriminatórios diretos, indiretos, ilegais ou abusivos; e

VI – direito à privacidade e à proteção de dados pessoais, nos termos da legislação pertinente.

Parágrafo único. Os agentes de inteligência artificial informarão, de forma clara e facilmente acessível, os procedimentos necessários para o exercício dos direitos descritos no *caput*.

Se um modelo de IA atinge **99% de acurácia** em um teste de laboratório, **mas não consegue explicar o porquê de suas decisões**, podemos dizer que ele é "inteligente" o suficiente para ser implementado em **decisões que afetam vidas humanas**?

Se um modelo de IA atinge **99% de acurácia** em um teste de laboratório, **mas não consegue explicar o porquê de suas decisões**, podemos dizer que ele é "inteligente" o suficiente para ser implementado em **decisões que afetam vidas humanas**?

“the problem is that a single metric, such as classification accuracy, is an incomplete description of most real-world tasks’

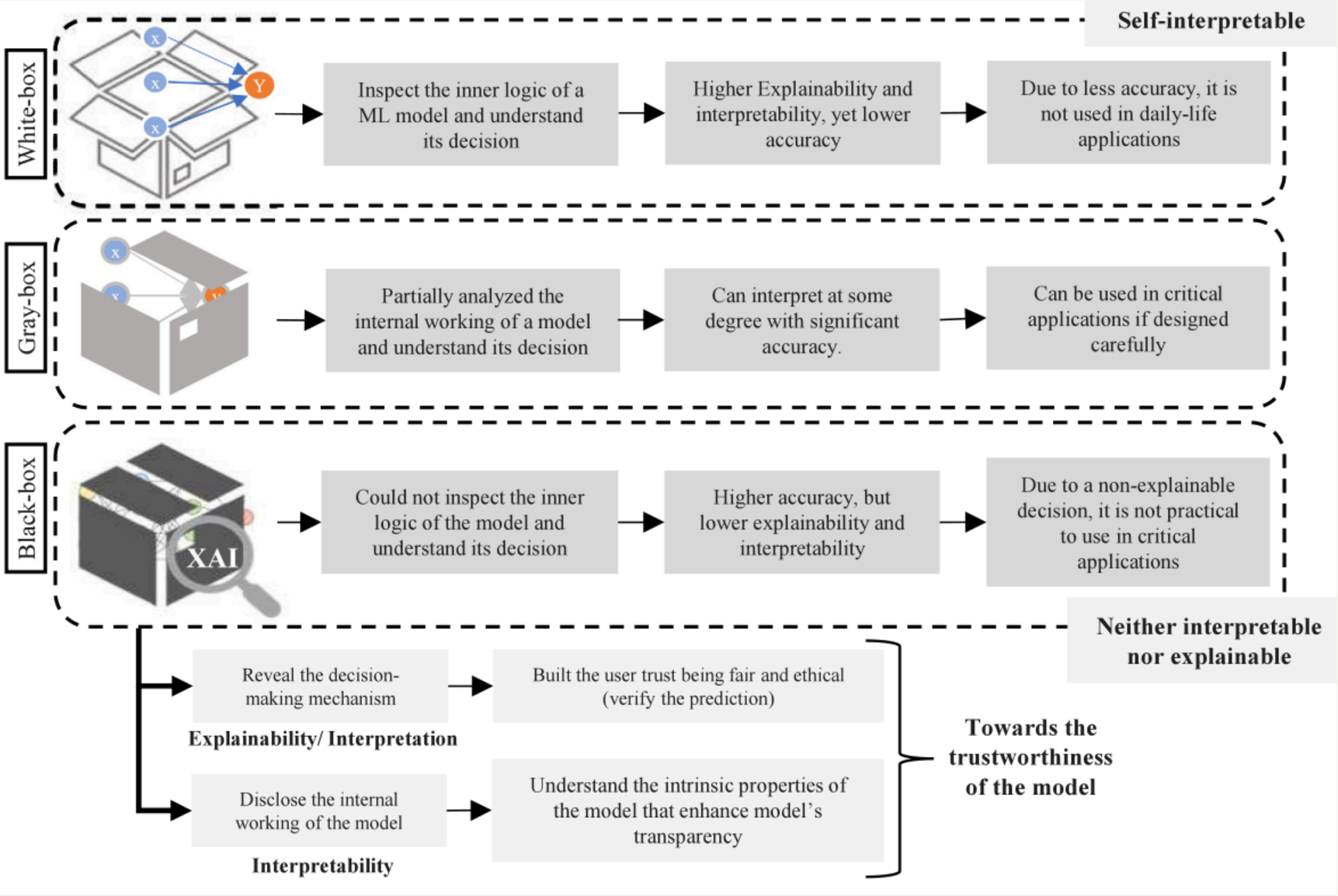
(Finale Doshi-Velez and Been Kim)

interpretabilidade

- voltada para desenvolvedores e especialistas;
- permite análise de como a saída do modelo foi construída.

explicabilidade

- voltada para o usuário final;
- justificar a decisão do modelo;



árvore de decisão

regressão logística

KNN

Naive Bayes

Redes Neurais

Métodos de Explicabilidade Agnósticos ao Modelo - LIME

Local Interpretable Model-agnostic Explanations (LIME)
Ribeiro et al., 2016

Algoritmo:

1. Escolha um exemplo que você quer explicar;
2. Gere 'perturbações' desse exemplo;
3. Realize a inferência de cada perturbação;
   a. registra um novo dataset (entrada perturbada localmente, saída)
4. Através de um kernel de proximidade, atribui maior peso aos exemplos mais próximos ao exemplo original;
5. Treina um modelo caixa-branca (regressão linear/LASSO);
6. Entrega os pesos da regressão: importância de cada feature;

Algorithm 1 Sparse Linear Explanations using LIME

Require: Classifier f, Number of samples N
Require: Instance x, and its interpretable version x'
Require: Similarity kernel π_x, Length of explanation K

$\mathcal{Z} \leftarrow \{\}$
for $i \in \{1, 2, 3, ..., N\}$ **do**
  $z_i' \leftarrow sample_around(x')$
  $\mathcal{Z} \leftarrow \mathcal{Z} \cup \langle z_i', f(z_i), \pi_x(z_i) \rangle$
end for
$w \leftarrow$ K-Lasso$(\mathcal{Z}, K)$ $\triangleright$ with z_i' as features, $f(z)$ as target
return w

Ribeiro, Marco Tulio, Sameer Singh, and Carlos Guestrin. "" Why should i trust you?" Explaining the predictions of any classifier." Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining. 2016.

Métodos de Explicabilidade Agnósticos ao Modelo - LIME

Local Interpretable Model-agnostic Explanations (LIME)
Ribeiro et al., 2016

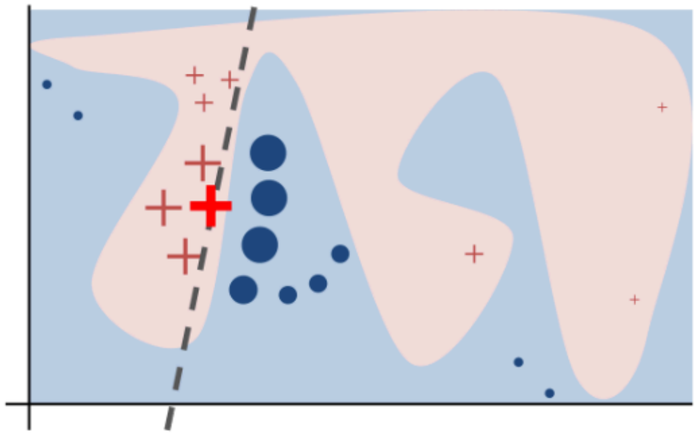
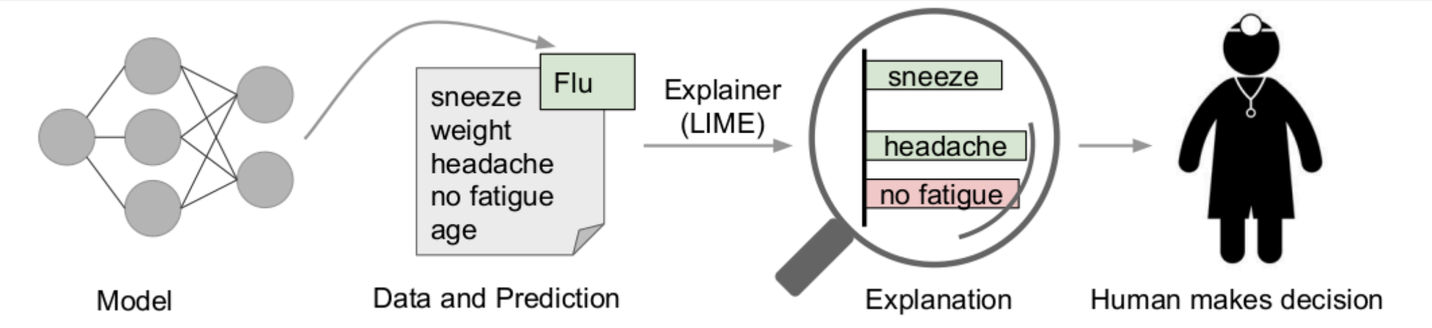


Figure 3: Toy example to present intuition for LIME. The black-box model's complex decision function f (unknown to LIME) is represented by the blue/pink background, which cannot be approximated well by a linear model. The bold red cross is the instance being explained. LIME samples instances, gets predictions using f, and weighs them by the proximity to the instance being explained (represented here by size). The dashed line is the learned explanation that is locally (but not globally) faithful.



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Métodos de Explicabilidade Agnósticos ao Modelo - SHAP

SHapley Additive exPlanations (SHAP)
Lundberg et al., 2017

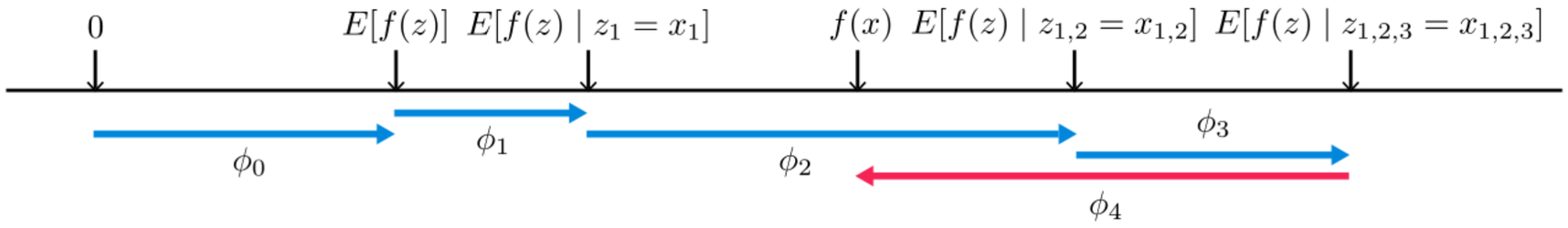

Figure 1: SHAP (SHapley Additive exPlanation) values attribute to each feature the change in the expected model prediction when conditioning on that feature. They explain how to get from the base value $E[f(z)]$ that would be predicted if we did not know any features to the current output $f(x)$. This diagram shows a single ordering. When the model is non-linear or the input features are not independent, however, the order in which features are added to the expectation matters, and the SHAP values arise from averaging the ϕ_i values across all possible orderings.

Lundberg, Scott M., and Su-In Lee. "A unified approach to interpreting model predictions." Advances in neural information processing systems 30 (2017).

Métodos de Explicabilidade Agnósticos ao Modelo - SHAP

SHapley Additive exPlanations (SHAP)
Lundberg et al., 2017

Blackbox model

Input datapoint

Age

$$\phi_i(f, x) = \sum_{z' \subseteq x'} \frac{|z'|!(M - |z'| - 1)!}{M!} [f_x(z') - f_x(z' \setminus i)]$$

Shapley value for feature i

Subset

Simplified data input

(para dados tabulares x'=x)

(z' é um subconjunto de features de x)

(o valor shapley é calculado considerando todos os subset de features z' em x)

$x =$ | Age = 56 | Gender = F | Body Mass Index = 30 | Heart disease = yes | ... |

Lundberg, Scott M., and Su-In Lee. "A unified approach to interpreting model predictions." Advances in neural information processing systems 30 (2017).

Métodos de Explicabilidade Agnósticos ao Modelo - SHAP

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(estamos procurando a diferença no output ao retirar a feature i do subset z')

Blackbox model

Input datapoint

Age = 56 | Body Mass Index = 30

Age

$$\phi_i(f, x) = \sum_{z' \subseteq x'} \frac{|z'|!(M - |z'| - 1)!}{M!} [f_x(z') - f_x(z' \setminus i)]$$

Shapley value for feature i

Subset

Simplified data input

Body Mass Index = 30

$x =$ Age = 56 | Gender = F | Body Mass Index = 30 | Heart disease = yes | ...

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Métodos de Explicabilidade Agnósticos ao Modelo - SHAP

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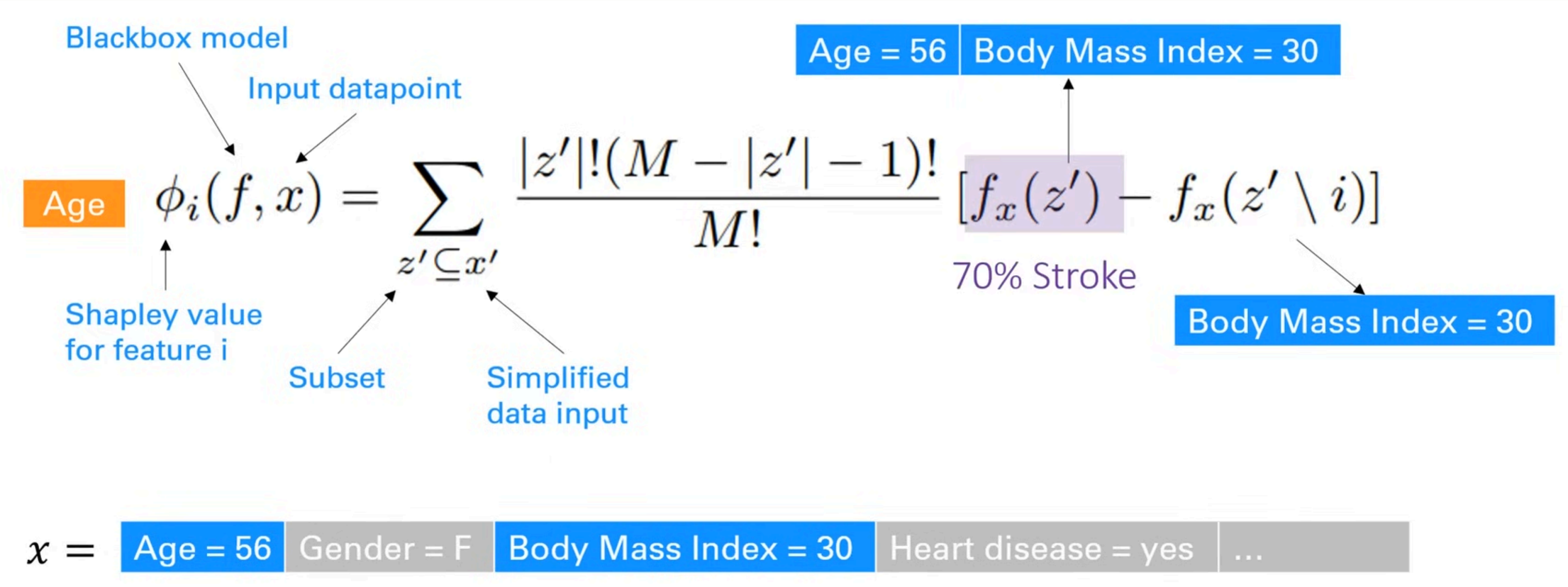
(quando considera o atributo 'idade', a probabilidade de 'derrame' é de 70%)



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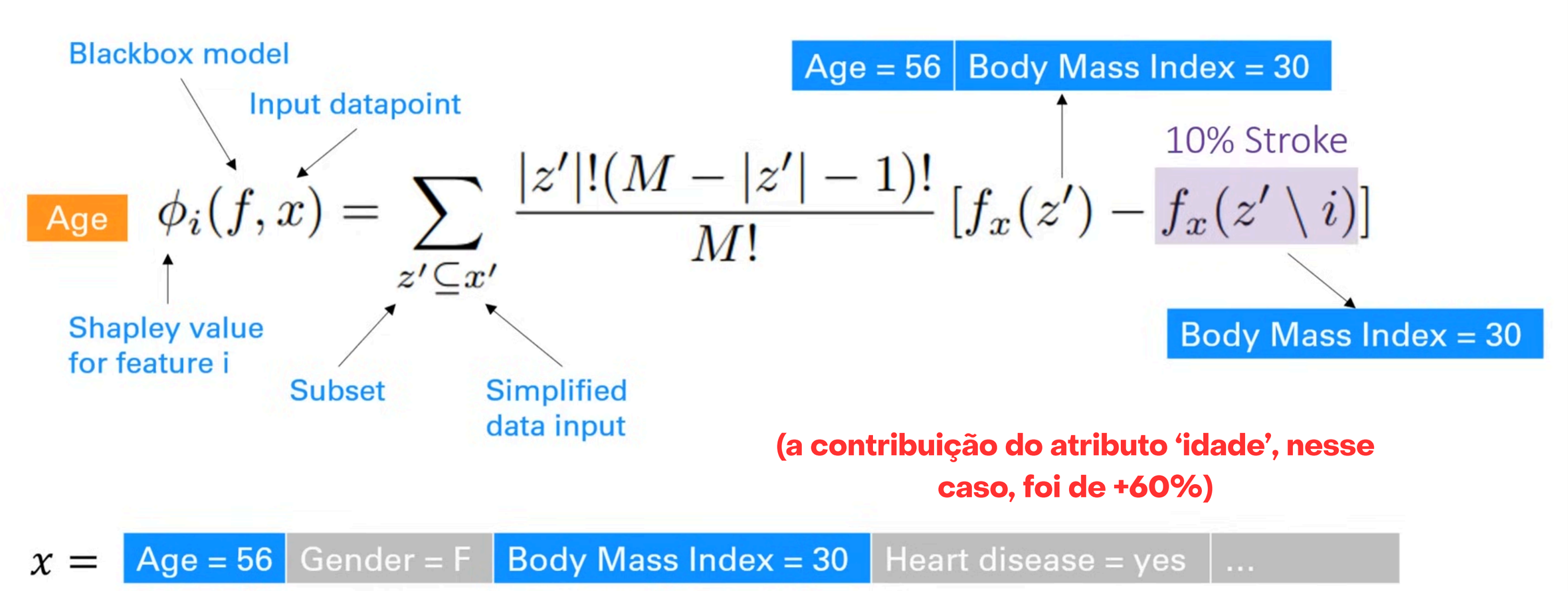
Métodos de Explicabilidade Agnósticos ao Modelo - SHAP

SHapley Additive exPlanations (SHAP)
Lundberg et al., 2017

(sem o atributo 'idade', a probabilidade de 'derrame' cai para 10%)



(a contribuição do atributo 'idade', nesse caso, foi de +60%)

Lundberg, Scott M., and Su-In Lee. "A unified approach to interpreting model predictions." Advances in neural information processing systems 30 (2017).

Métodos de Explicabilidade Agnósticos ao Modelo - SHAP

SHapley Additive exPlanations (SHAP)
Lundberg et al., 2017

(as contribuições são ponderadas considerando quantas features |z'| do total do total de features M foram usadas)

Blackbox model

Input datapoint

Age

$$\phi_i(f, x) = \sum_{z' \subseteq x'} \frac{|z'|!(M - |z'| - 1)!}{M!} [f_x(z') - f_x(z' \setminus i)]$$

Shapley value for feature i

Subset

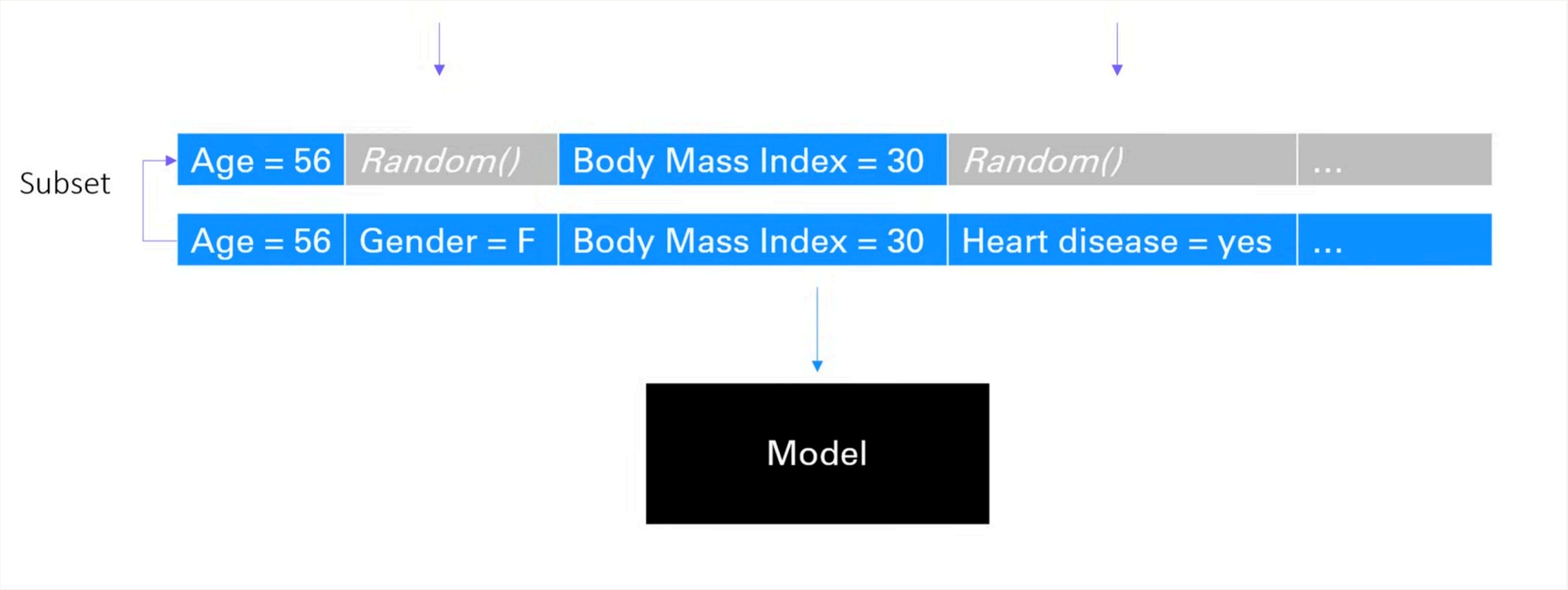
Simplified data input

Weighting

(a contribuição de atributo 'age' deve pesar mais em um subset onde já existam muitas features)

$x =$ | Age = 56 | Gender = F | Body Mass Index = 30 | Heart disease = yes | ... |

Lundberg, Scott M., and Su-In Lee. "A unified approach to interpreting model predictions." Advances in neural information processing systems 30 (2017).



(avaliar um modelo apenas com um subconjunto de features, é equivalente a randomizar os valores das features que devem ser apagadas)

(não existe retreinamento, a explicabilidade desejada parte de um modelo treinado e fixado)

Métodos de Explicabilidade Agnósticos ao Modelo - SHAP

SHapley Additive exPlanations (SHAP)
Lundberg et al., 2017

$$f(x) \;\approx\; \phi_0 \;+\; \sum_{i=1}^{d} \phi_i$$

$$\phi_0 \;=\; \mathbb{E}[f(X)] \;\approx\; \frac{1}{M}\sum_{m=1}^{M} f\Big(x^{(m)}\Big), \qquad x^{(m)} \sim X$$

$$\phi_i \;=\; \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(d-|S|-1)!}{d!}\Big(v(S \cup \{i\}) - v(S)\Big), \qquad N = \{1,\ldots,d\}.$$

Lundberg, Scott M., and Su-In Lee. "A unified approach to interpreting model predictions." Advances in neural information processing systems 30 (2017).

Projeto da Disciplina - Tema 1

Usando a metodologia descrita em "Enhancing the Interpretability of SHAP Values Using Large Language Models"[1]

Executando SLMs localmente usando o Docker Model Runner[2]

Avalie:

1.1 Usando uma SLM fixa (e.g. qwen2.5, ibm granite 4.0 nano)[3], dada uma nova query x, estude como escolher de maneira adptativa o número de perturbações necessárias do LIME para obter uma explicabilidade coerente de x via texto;

1.2 Usando uma SLM fixa, uma explicação 'pobre' é aquela gerada com custo computacional reduzido no LIME ou SHAP (menor número de inferências). A partir de um número grande de N explicações 'pobres' via texto, é possível resumi-las (usando outra LLM) em uma única explicação forte?

1 - https://arxiv.org/pdf/2409.00079
2 - https://www.youtube.com/watch?v=GOgfQxDPaDw
3 - hub.docker.com/r/ai/qwen2.5, hub.docker.com/r/ai/granite-4.0-h-nano